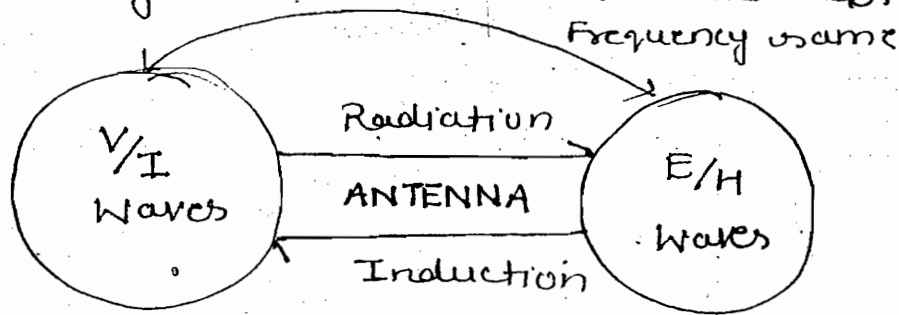


ANTENNAS :-

- Hertzian Dipole / Halfwave Dipole
- Basic Terms and Definitions → 50%
- Antenna Arrays
- Friis - Free Space Propagation
- Study / Classification of Antennas



Hertzian Dipole as a Basic Radiating Element :-

It is a dl length $I(t) = I_m \sin \omega t$ (oscillatory or Harmonic) current element with $dl \ll \lambda$

$$(i) \quad A(t) = \frac{\mu I(t) dl}{4\pi r}$$

$$(ii) \quad B(t) = \mu H(t) = \nabla \times \vec{A}$$

$$(iii) \quad \nabla \times H = \epsilon \frac{\partial E}{\partial t}$$

$$(iv) \quad E(t) = \frac{1}{\epsilon} \int (\nabla \times H) dt$$

Every harmonic current produces a time/space harmonic E & H around it. This EM wave generation is called as radiation.

Expression for Radiated fields of Hertzian Dipole :-

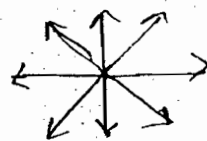
$$\left. \begin{array}{l} \text{Radiated} \\ \text{Wave} \end{array} \right\} \begin{array}{l} E(r, \theta, \phi, t)_{\theta} = \left(\frac{I_m dl \sin \theta \cdot \omega}{4\pi \epsilon c^2 r} \right) \sin \omega t \cdot e^{-j\beta r} a_{\theta} \\ H(r, \theta, \phi, t)_{\phi} = \left(\frac{I_m dl \sin \theta \omega}{4\pi c r} \right) \sin \omega t \cdot e^{-j\beta r} a_{\phi} \end{array}$$

$$W \rightarrow E(z,t)_x = E_0 e^{j\omega t} e^{-\gamma z} a_x$$

Properties of Radiated Waves:-

- i) They travel radially outward and hence their amplitudes decrease as $\frac{1}{r}$ due to their power density decreases as $\frac{1}{r^2}$

→ Amplitude dec. but does not attenuate.



$$E_\theta \times H_\phi = P_r$$

$$\frac{E_\theta}{H_\phi} = \frac{1}{\epsilon c} = \frac{1}{\epsilon \sqrt{\frac{1}{\mu \epsilon}}} = \sqrt{\frac{\mu}{\epsilon}} = 120\pi \Omega$$

- (ii) The amplitude of the radiated wave is not omni-directional and depends on θ and ϕ i.e. Radiation is directive in nature.

- (iii) The amplitude of the radiated wave always depends on $\left(\frac{dl}{\lambda}\right)$ ratio i.e. frequency decides radiated power

Total Power Radiated from a Hertzian Dipole:-

$$W_r = \oint P_{avg} \cdot ds$$

$$= \oint_{\text{Sphere}} \frac{1}{2} \frac{E_0^2}{\eta} a_r \cdot ds \cdot a_r$$

$$W_r = \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \frac{1}{2} \left(\frac{I_m d \sin \theta \omega}{4\pi \epsilon c^2 r} \right)^2 \frac{1}{120\pi} r^2 \sin \theta d\theta d\phi$$

$$\left(\omega = \frac{2\pi c}{\lambda} \right)$$

$$W_r = I_{rms}^2 \cdot 80 \pi^2 \left(\frac{dl}{\lambda} \right)^2$$

The expression is similar to a resistor dissipating power as heat i.e. every antenna radiates power in the form of EM wave

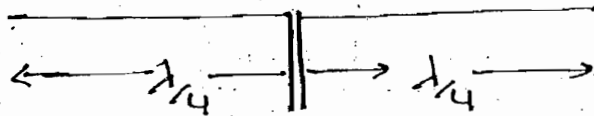
$$R_r = \text{Radiation resistance of Hertzian Dipole}$$

$$= 80 \pi^2 \left(\frac{dl}{\lambda} \right)^2$$

→ R_r is a measure of radiated power for a given input current i.e. it should be as possible for a practical antenna

Halfwave dipole as a fundamental Antenna! -

→ A centre feed $\lambda/2$ - dipole is a transmission opened out by $\lambda/4$ on either sides



→ The length of antenna is strictly depends on the frequency of operation

FM

$$f = 100 \text{ MHz}$$

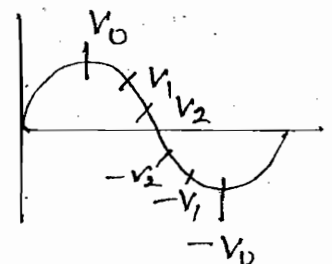
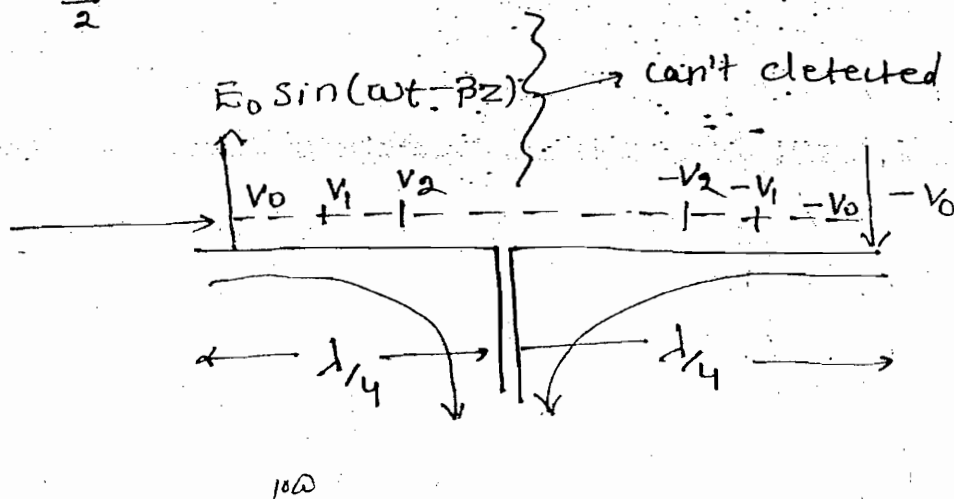
$$\lambda = 3 \text{ m}$$

$$\frac{\lambda}{2} = 1.5 \text{ m}$$

GSM

$$f = 1800 \text{ MHz}$$

$$\lambda = \text{few cm}$$



Note:-

When an EM waves having right frequency and right polarization travels along the axis of the antenna, it induced V_0 voltage at one end and progressively decrease voltage such that it is $-V_0$ at the other end. Hence the word half wave dipole.

> These voltages drive a current towards centre which is maximum in the line such that

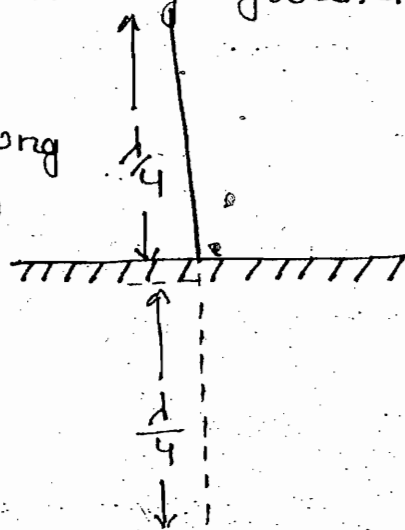
$$\text{Asymptotic } \left\{ \begin{array}{l} |V(z)| = V_0 \sin \beta z \\ |I(z)| = I_0 \cos \beta z \end{array} \right\} \rightarrow z = -\frac{\lambda}{4} \text{ to } \frac{\lambda}{4}$$

$\frac{V}{I}$
distribution

* $V_0 = -ve$ at other end bec. of phase diff = 180°

Quarterwave Monopole as a Practical Antenna:-

- It is a low frequency vertically grounded single wire of $\lambda/4$ length
- It is a base feed and along with its image works like a harmonic dipole.



Summary:-

A halfwave dipole is an array of Hertzian dipoles with dl from $-\lambda/4$ to $\lambda/4$ and

$$\underline{I_m = |I(z)| = I_0 \cos \beta z}$$

Asymptotic

Radiation Expressions for Halfwave Dipole:-

$$E(r, \theta, \phi, t)_{\theta} = \left(\frac{60 I_0}{r} \frac{\cos\left(\frac{\pi}{2} \cos\theta\right)}{\sin\theta} \right) \sin\omega t \cdot e^{-j\beta r} a_{\theta}$$

$$H(r, \theta, \phi, t)_{\phi} = \left(\frac{I_0}{2\pi r} \frac{\cos\left(\frac{\pi}{2} \cos\theta\right)}{\sin\theta} \right) \sin\omega t \cdot e^{-j\beta r} a_{\phi}$$

Total power radiated from Halfwave Dipole

$$W_r = \int \frac{1}{2} \frac{E_0^2}{\eta} \cdot dV$$

$$= \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \frac{1}{2} \left(\frac{60 I_0}{r} \frac{\cos\left(\frac{\pi}{2} \cos\theta\right)}{\sin\theta} \right)^2 \cdot \frac{1}{120\pi} r^2 \sin\theta d\theta d\phi$$

$$= I_{rms}^2 (73)$$

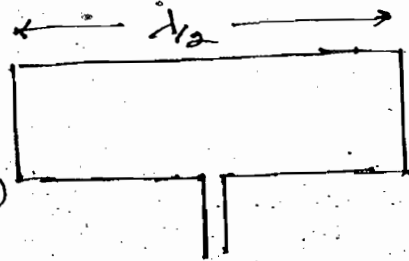
R_r for a Halfwave Dipole = 73Ω

R_r for Quarterwave monopole = 36.5Ω

R_r for a folded dipole = $2^2 \times 73 \Omega = 292 \Omega$

R_r for n -Dipoles = $n^2 \times 73$

($n^2 \times 73$)



Basic Terms and Definitions:-

1) Isotropic Antenna:-

It radiates power in all directions uniformly. Its E field pattern is independent of θ and ϕ

eg:- Broadcast antenna

(II) Radiation Power Density :-

It is the strength of the radiated EM wave in any direction at any distance from the antenna

$$\begin{aligned}\frac{\text{Power}}{\text{Area}} \text{ or Watts/m}^2 &= \frac{dW_r}{ds} \\ &= \text{Poynting vector of EM wave} \\ &= \frac{1}{2} \frac{E_0^2}{\eta} (r, \theta, \phi) \\ &= V(r, \theta, \phi)\end{aligned}$$

mp (III) Radiation Power Intensity :-

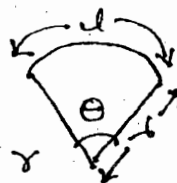
It is the strength of the radiated EM wave in any direction from the antenna

$$\begin{aligned}\frac{\text{Power}}{\text{direction}} &= \frac{\text{Power/solid angle}}{\text{}} \\ &= \text{with steradian} = \frac{dW_r}{d\Omega}\end{aligned}$$

$$d = \theta r$$

If $\theta = 1$ then $d = r$

If $\theta = 6.28$ radian then $C = 6.28 r$

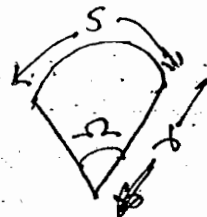


$$S = 2\pi r^2$$

$$\Omega = 1 \text{ steradian}$$

$$S = r^2$$

$$\Omega = 12.56 \text{ steradian}$$



$$\text{Total surface area} = 12.56 r^2$$

Any small incremental area, $ds = r^2 d\Omega = r^2 \sin\theta \cdot d\theta \cdot d\phi$
 $d\Omega = \sin\theta d\theta d\phi$

$$\frac{dW_r}{d\Omega} = \frac{dW_r}{ds} \cdot \frac{ds}{d\Omega}$$

$$= V(r, \theta, \phi) r^2 = \Psi(\theta, \phi)$$

eg:- Isotropic Antenna

$$\frac{dW_r}{ds} = \frac{W_r}{4\pi r^2} = V_{avg}$$

$$\frac{dW_r}{d\Omega} = \frac{W_r}{4\pi} = \Psi_{avg}$$

IV) Gain of an Antenna:-

$G_D \rightarrow$ Directive gain

$G_P \rightarrow$ Power gain

$D \rightarrow$ Directivity

(a) Directive Gain (G_D):-

The radiation intensity of the antenna in a given direction to the radiation intensity of isotropic antenna

$$G_D = \frac{\Psi(\theta, \phi)}{\Psi_{avg}} = \frac{4\pi \Psi(\theta, \phi)}{W_r} = \frac{4\pi \Psi(\theta, \phi)}{\int \Psi(\theta, \phi) d\Omega}$$

$G_D > 1$ or $G_D < 1 \rightarrow$ depends on direction

(b) Power Gain:-

$$G_P = \frac{4\pi \cdot \Psi(\theta, \phi)}{W_N}$$

$W_R = \text{Total o/p power}$

$W_N = \text{Total i/p power}$

$$G_p = \frac{4\pi \psi(\theta, \phi)}{W_R} \cdot \frac{W_R}{W_N}$$

$$= G_p \times \text{Efficiency of Radiation}$$

$$\text{Efficiency} = \frac{W_R}{W_R + W_L} = \frac{R_r}{R_r + R_l}$$

$$\text{Directivity} = G_p|_{\max}$$

$$\boxed{D \geq 1} \rightarrow \text{Always}$$

$D = 1$ for isotropic antenna

Radiation Pattern of an Antenna:-

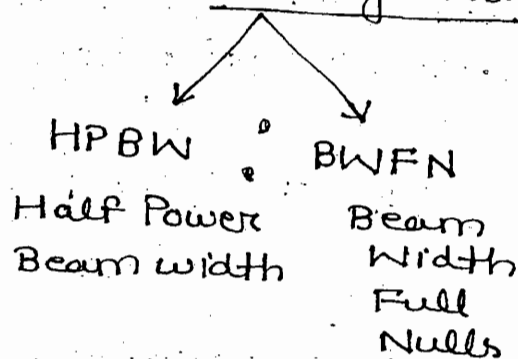
It is a polar plot of radiation intensity indicating various directions around the antenna where radiation is finitely strong

eg:- $|E| = \frac{k \sin \theta}{r}$

$$\psi = \psi_0 \sin^2 \theta$$

$$\theta = 0^\circ / 180^\circ$$

$$\left\{ \begin{array}{l} |E| = 0 \rightarrow \text{Null points} \\ \rightarrow \theta_{NP} \end{array} \right.$$



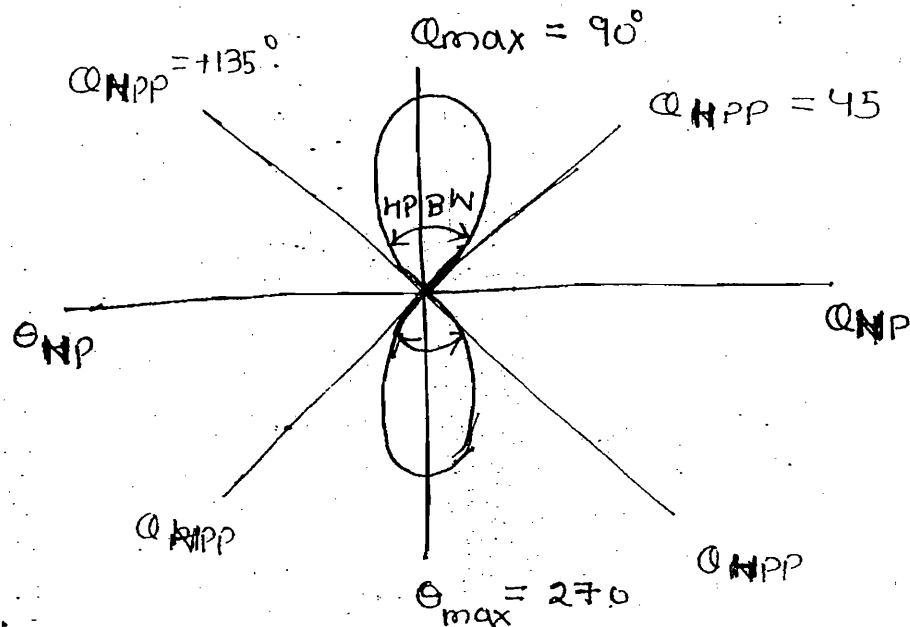
$$\theta = 90^\circ / 270^\circ$$

$$|E| = E_{\max}$$

$$\theta_{\max}$$

$$\theta = 45^\circ / 135^\circ / 225^\circ / 315^\circ$$

$$|E| = \frac{E_{\max}}{\sqrt{2}}$$



$$\begin{aligned} \theta_{\text{HPBW}} &= \text{HPP is next HPP in the maxima} \quad (\text{Half power pt.}) \\ &= 135^\circ - 45^\circ = 90^\circ \end{aligned}$$

The antenna considered has a ϕ independent pattern and hence a circular top view of the beam (for all ϕ)

In general

$$\begin{aligned} \theta_{\text{HPBW}} \times \phi_{\text{HPBW}} &= \Omega_A \text{ Beam solid angle} \\ &= \text{steradian} = \text{radian}^2 \end{aligned}$$

In circular

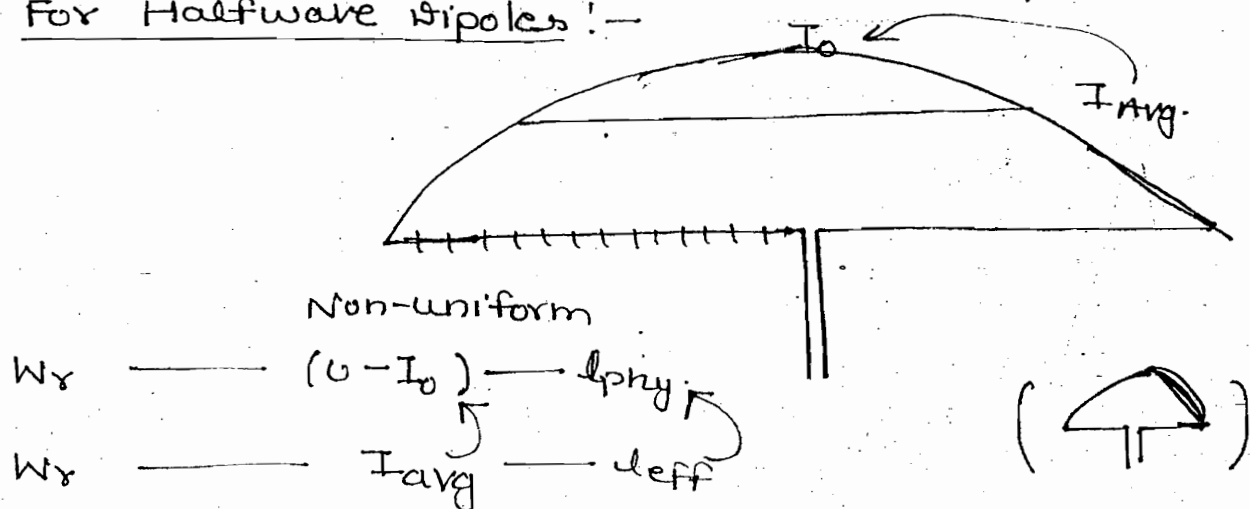
$$\left(\theta_{\text{HPBW}} \right)^2 = \Omega_A$$

$$\Omega \propto \frac{1}{\Omega_A}$$

$$\Rightarrow \boxed{\Omega = \frac{4\pi}{\Omega_A}}$$

Q (VI) Effective length of an antenna:-

(a) For Halfwave dipoles:-



Note:-

→ An antenna radiates W_r power over its physical length and non-uniform currents everywhere. Effective length is a length required to radiate same power assuming uniform currents everywhere.

$$I_{avg} = \frac{1}{\pi} \int_0^{\pi} I_0 \sin \theta d\theta = \frac{2I_0}{\pi}$$

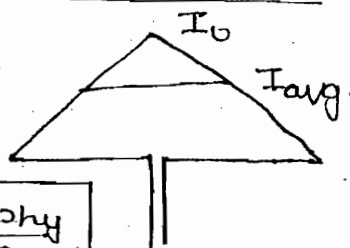
$$\boxed{l_{eff} = \frac{2l_{phy}}{\pi}}$$

(b) For electrically short dipoles ($l_{phy} < \frac{\lambda}{10}$):-

$|I(z)| \propto z$
linear current

$$I_{avg} = \frac{I_0}{2}$$

$$\boxed{l_{eff} = \frac{l_{phy}}{2}}$$



c) For Hertzian Dipole ($l_{\text{phy}} < \frac{\lambda}{25}$) :-

$$l_{\text{phy}} = l_{\text{eff}} = dl$$

$$I_{\text{avg}} = I_m \rightarrow \text{Uniform currents}$$

$$\text{For Hertzian Dipole } R_r = 80\pi^2 \left(\frac{dl}{\lambda} \right)^2$$

$$\text{For any Dipole, } R_r = 80\pi^2 \left(\frac{l_{\text{eff}}}{\lambda} \right)^2$$

$$\begin{aligned} \text{For Electrically short Dipoles } R_r &= 80\pi^2 \left(\frac{l}{2\lambda} \right)^2 \\ &= 20\pi^2 \left(\frac{l}{\lambda} \right)^2 \end{aligned}$$

$$\text{" " Monopoles } R_r = 10\pi^2 \left(\frac{l}{\lambda} \right)^2$$

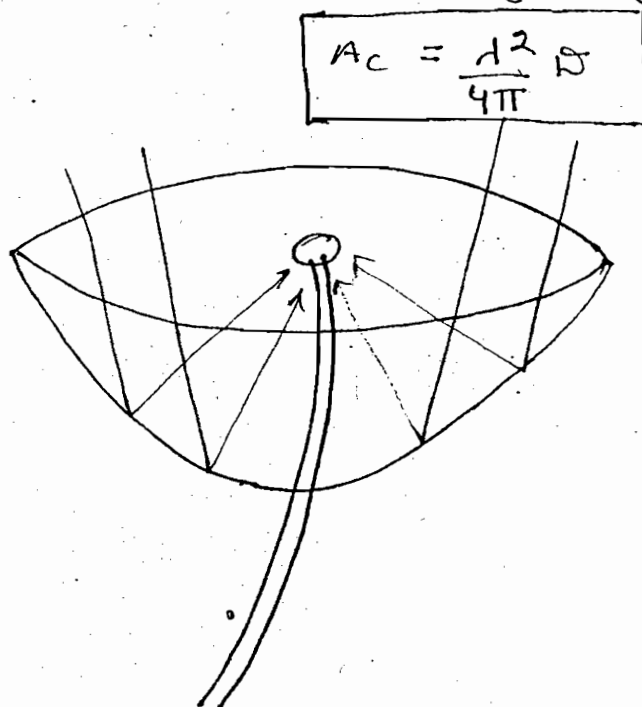
$$\begin{aligned} \text{Monopoles over conducting earth} \quad R_r &= 10\pi^2 \left(\frac{2h}{\lambda} \right)^2 \\ &= 40\pi^2 \left(\frac{h}{\lambda} \right)^2 \end{aligned}$$

$$\begin{aligned} \text{For halfwave dipole } &= 80\pi^2 \left(\frac{2l}{\pi\lambda} \right)^2 \\ &= 320\pi^2 \left(\frac{l}{\lambda} \right)^2 = 80\Omega \end{aligned}$$

VII) Capture Area or Effecting Area (A_c) :-

In microwave antenna $l = \text{few cm}$
 $r = \text{few } 1000\text{km}$

$$A_c = \text{Capture Area} = \frac{\text{Power induced}}{\text{Poynting Vector}}$$



Workbook:-

$$\theta \rightarrow G_{1\theta} |_{\max} \rightarrow \Psi(\theta, \phi) \rightarrow U(r, \theta, \phi) \rightarrow E(r, \theta, \phi)$$

$$\text{Hertzian Dipole, } |E| = \frac{k \sin \theta}{r}$$

$$U(r, \theta, \phi) = \frac{1}{2} \frac{k^2 \sin^2 \theta}{r^2 \cdot \eta}$$

$$\Psi(\theta, \phi) = \frac{1}{2} \frac{k^2 \sin^2 \theta}{\eta}$$

$$G_{1\theta} = 4\pi^2 \frac{1}{2} \frac{k^2 \sin^2 \theta}{\eta}$$

$$\int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \frac{1}{2} \frac{k^2 \sin^2 \theta}{\eta} \sin \theta \, d\theta \cdot d\phi = \underline{2 \sin^2 \theta}$$

$$\int_{\theta=0}^{\pi} \sin^3 \theta \, d\theta = \frac{4}{3}$$

$\Leftarrow$

$$= \frac{3}{2} \sin^2 \theta$$

$$D = G D_{\max} = 1.5$$

For Halfwave Dipole, $D = 1.63$

$4 - \frac{\lambda}{2}$ Dipole

$$W_r = 4 I_{rms}^2 \cdot 73 = 4 \left(\frac{0.5}{\sqrt{2}} \right)^2 \cdot 73 = 36.5 \text{ Watts}$$

$$D = \frac{4\pi}{\Omega_A}$$

$$10 \log D = 44 \text{ dB}$$

$$\Rightarrow D = \frac{4\pi}{(\theta_{HPBW})^2}$$

$$\Rightarrow D = 10^{4.4}$$

$$10^{4.4} = \frac{4 \times 3.14 \times (57)^2}{(\theta_{HPBW})^2}$$

$$\text{Efficiency} = \frac{R_r}{R_r + R_d}$$

R_r for $\lambda/8$ Dipole

$$R_r \quad \lambda/2 \quad \text{---} \quad 73 \Omega$$

$$R_r \quad \lambda/4 \quad \text{---} \quad 36.5 \Omega$$

$$R_r \quad \lambda/8 \quad \text{---} \quad 18.25 \Omega$$

$$\text{Efficiency} = \frac{18.25}{19.75} = 89\%$$

$$\begin{aligned} \lambda &\rightarrow 492 \text{ m} \\ d &\rightarrow 124 \text{ m} \\ \frac{\lambda}{4} &= 123 \text{ m} \end{aligned}$$

$$R_r = 36.5 \Omega$$

$$10 \log 4 = 6 \text{ dB}$$

$$G = 4$$

$$W_{IN} = 1 \text{ mW}$$

lossless

$$W_r = W_{IN}$$

→ Ans - (B)

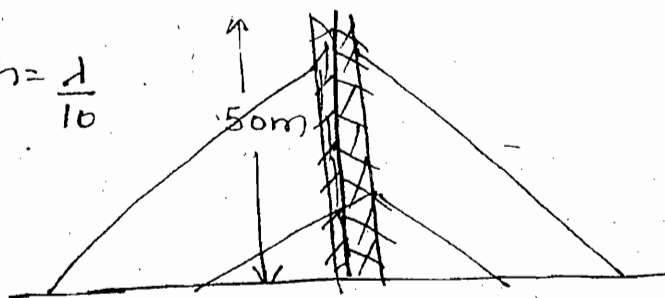
(B)

$$h = 50 \text{ m}$$

$$f = 60 \text{ KHz}$$

$$\lambda = \frac{3 \times 10^8}{6 \times 10^5} = 500 \text{ m}$$

$$h = \frac{\lambda}{10}$$



$$R_r = 40 \pi^2 \left(\frac{50}{500} \right)^2$$

$$\approx 4 \Omega = \frac{2 \pi^2}{5}$$

$$125 \text{ — } \frac{\lambda}{4} \text{ — } 36.5 \Omega$$

$$50 \text{ m — } \frac{\lambda}{10} \text{ — } 4 \Omega$$

$$W_r = I_{rms}^2 R_r$$

$$\Psi_{avg} = \frac{100}{4 \pi} = \frac{25}{\pi} = 7.96$$

$$V_{avg} = \frac{100}{4 \pi \times (10 \times 10^3)^2} = 7.96 \times 10^{-8}$$

$$= 0.08 \mu \text{ W}$$

$$\Psi \propto W_r \propto R_r \propto \frac{1}{\lambda^2} \propto f^2$$

strongly

$$f' = f/2$$

$$\Psi' = \Psi/4$$

$$\eta = \rho \quad \eta = \rho \text{ dB}$$

$$\eta = \frac{4 \pi \Psi(\theta, \phi)_{max}}{W_r}$$

$$= \frac{4 \pi \cdot 150}{0.9 (W_{in})} = \frac{4 \pi \cdot 150}{36 \pi} = 16.67 = \eta$$

$$10 \log 16.67 = 11.76 \text{ dB}$$

$$\eta = 11.76 \text{ dB}$$

1A - 1m - using

$\rightarrow I_{rms}$

$$f = 10 \text{ MHz}$$

$$\lambda = \frac{3 \times 10^8}{10 \times 10^6} \times 30 \text{ m}$$

$$l = \frac{\lambda}{30}$$

$$W_r = I_{rms}^2 R_r$$

$$R_r = 80 \pi^2 \left(\frac{1}{30} \right)^2 = 0.88 \Omega$$

$$W_r = 0.88$$

$$\left(\frac{dl}{\lambda} \right) \begin{cases} \frac{1}{2} \rightarrow 73 \Omega \\ \frac{1}{4} \rightarrow 36.5 \Omega \\ \frac{1}{10} \rightarrow 4 \Omega \\ \frac{1}{30} \rightarrow 0.88 \Omega \end{cases}$$

13. $\Theta_{HPBW} = 90^\circ$

14. A

15. Uniform currents

$$R_r = 80 \pi^2 \left(\frac{dl}{\lambda} \right)^2$$

$$l = 5 \text{ m}$$

$$\lambda = 100 \text{ m}$$

$$= 80 \pi^2 \left(\frac{1}{20} \right)^2 \approx 2 \Omega$$

16. A

$$l = 1.5 \text{ m}$$

$$f = 100 \text{ MHz}$$

$$\lambda = 3 \text{ m}$$

$$l = \frac{\lambda}{2}$$

$$l = 1.5 \text{ m}$$

$$f = 10 \text{ MHz}$$

$$\lambda = 30 \text{ m}$$

$$l = \frac{\lambda}{2}$$

Halfwave Dipole

ANTENNA ARRAYS! -

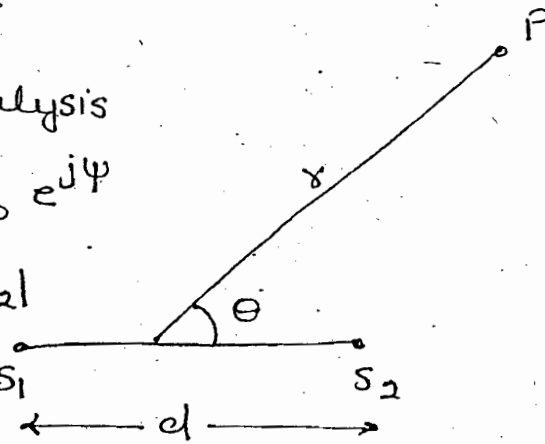
2-Element Isotropic Array! -

$r \gg d$ Far-zone analysis

$$E_T = E_1 + E_2 = E_0 + k E_0 e^{j\psi}$$

where $k=1$ if $|I_1| = |I_2|$

ψ = Phase diff. b/w the two fields



= current having phase difference (α) + path difference, giving phase

$$\text{Path difference} = d \cos \theta$$

$$\text{phase diff.} = \frac{2\pi}{\lambda} d \cos \theta = \beta d \cos \theta$$

$$\boxed{\psi = \alpha + \beta d \cos \theta}$$

$$E_T = E_0 (1 + \cos \psi + j \sin \psi)$$

$$|E_T| = |E_0| \sqrt{(1 + \cos \psi)^2 + \sin^2 \psi}$$

$$= |E_0| \sqrt{2 + 2 \cos \psi}$$

$$|E_T| = 2 E_0 \cos (\psi/2)$$

Case - (1)! -

$$d = \frac{\lambda}{2}, \quad \alpha = 0$$

$$\psi = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{2} \cos \theta$$

$$|E_T| = 2 E_0 \cos \left(\frac{\pi}{2} \cos \theta \right)$$

If $\theta = 90^\circ / 270^\circ \rightarrow \theta_{\max}$

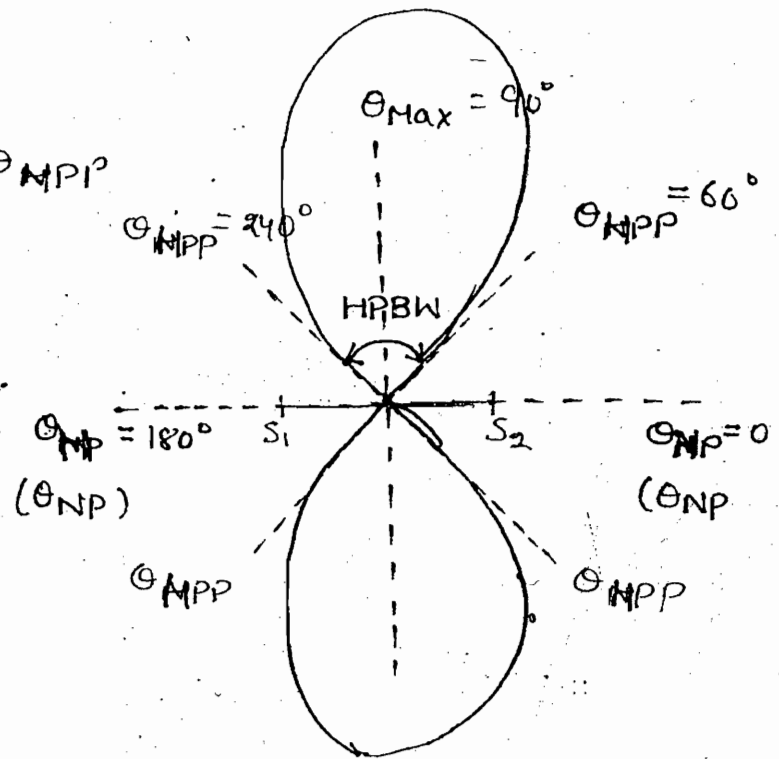
$$|E_T| = 2 E_0 = E_{\max}$$

If $\theta = 0^\circ / 180^\circ \rightarrow \theta_{NP}$

$$|E_T| = 0$$

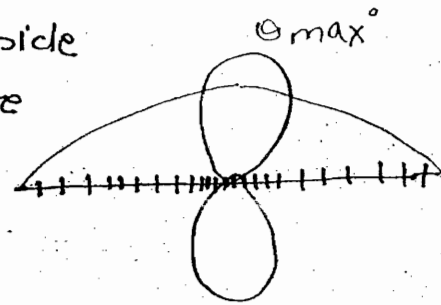
If $\theta = 60^\circ / 120^\circ / 240^\circ / 300^\circ \rightarrow \theta_{MPP}$

$$|E_T| = \frac{E_{max}}{\sqrt{2}}$$



Note: —

The currents on either side inphase and max. at the centre and halfwave dipole is also broad side pattern



Case - (ii) :-

$$d = \frac{\lambda}{2}, \quad \angle = \pi$$

$$\psi = \pi + \frac{2\pi}{\lambda} \cdot \frac{\lambda}{2} \cos \theta$$

$$|E_T| = 2E_0 \sin\left(\frac{\pi}{2} \cos \theta\right)$$

If $\theta = 90^\circ / 270^\circ \rightarrow \theta_{NP}$

$$|E_T| = 0$$

If $\theta = 0^\circ / 180^\circ \rightarrow \theta_{max}$

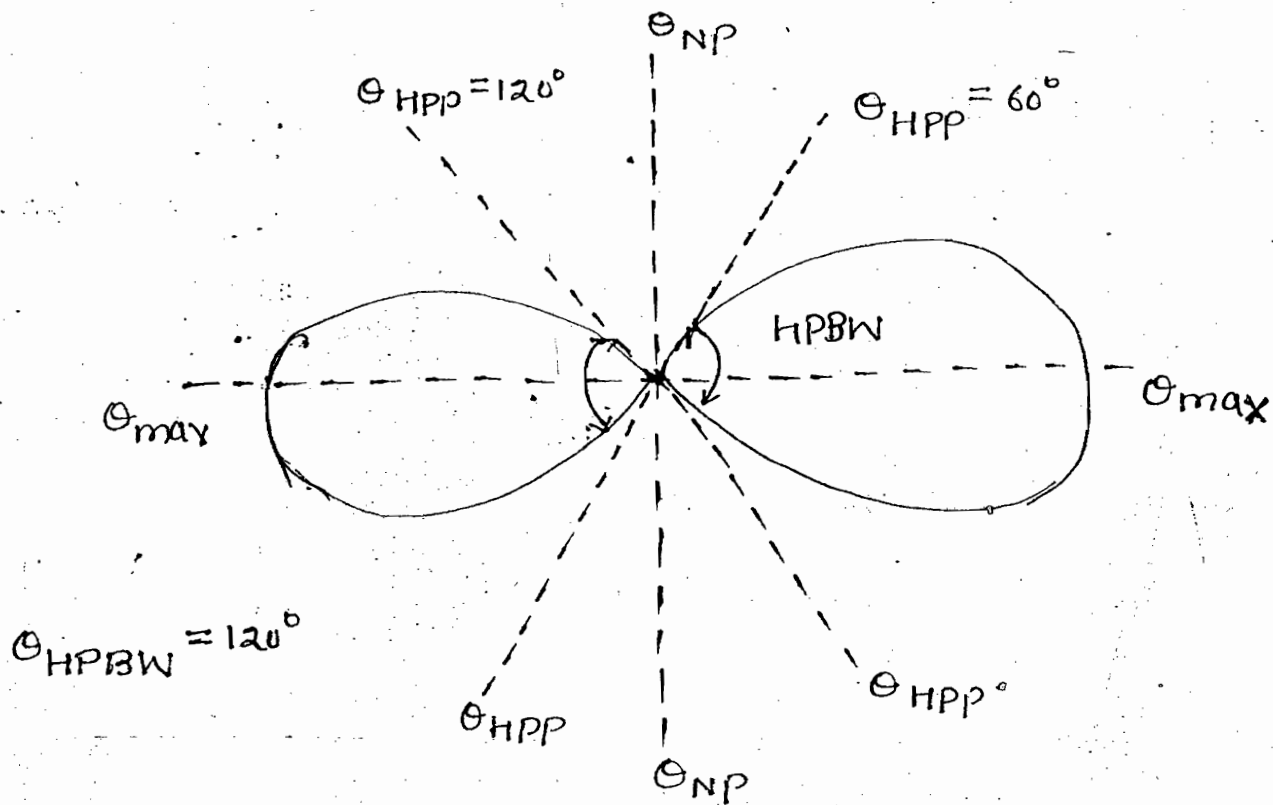
$$|E_T| = E_{max} = 2E_0$$

If $\theta = 60^\circ / 120^\circ / 240^\circ / 300^\circ \rightarrow \theta_{HPP}$

$$|E_T| = \frac{E_{max}}{\sqrt{2}}$$

This is called as End-fire Array whose

$$\theta_{max} = 0^\circ/180^\circ$$



Case - (III) :-

$$d = \lambda, \alpha = 0^\circ$$

$$E_T = 2E_0 \cos(\pi \cos \theta)$$

$$\text{If } \theta = 0^\circ/90^\circ/180^\circ/270^\circ$$

$$\longrightarrow \theta_{max}$$

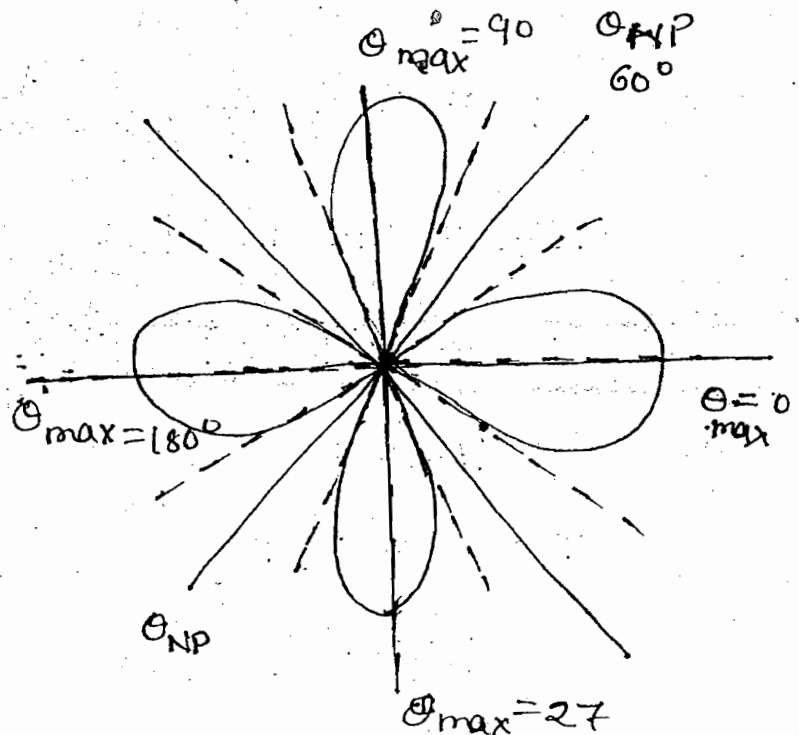
$$\text{If } \theta = 0^\circ/90^\circ/180^\circ/270^\circ$$

$$\longrightarrow \theta_{max}$$

$$\text{If } \theta = 60^\circ/120^\circ/240^\circ/300^\circ$$

$$\longrightarrow \theta_{NP}$$

$$\cos \theta = \pm \frac{1}{4} \text{ or } \pm \frac{3}{4}$$



In general $\theta_{\max}$ can be designed towards a specific direction

With $\psi \rightarrow 0$

$$\alpha + \beta d \cos \theta_{\max} = 0$$

$$\cos \theta_{\max} = \frac{-\alpha}{\beta d}$$

For broadside array, $\leftarrow$

$$\theta_{\max} = 90^\circ / 270^\circ$$

$$\Rightarrow \boxed{\alpha = 0}$$

In phase currents

For end-fire array

$$\theta_{\max} = 0^\circ / 180^\circ$$

$$\Rightarrow \boxed{\alpha = \pm \beta d}$$

$\theta_{\max}$

$\alpha = \pm$

Extension:-

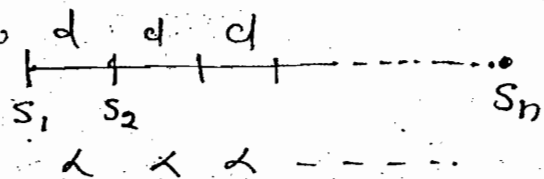
For n elements, uniform, linear isotropic sources

$$E_T = E_0 (1 + e^{j\psi} + e^{j2\psi} + \dots + e^{j(n-1)\psi})$$

G.P with $a=1$ $r=e^{j\psi}$

$$E_T = \frac{E_0 (1 - e^{jn\psi})}{(1 - e^{j\psi})}$$

$$\boxed{|E_T| = \frac{E_0 \sin(n\psi/2)}{\sin(\psi/2)}}$$



If $\psi \rightarrow 0$

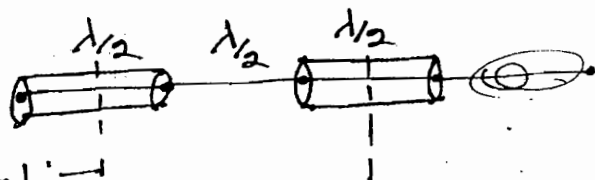
$$|E_T| = E_0 \frac{n(\psi/2)}{\psi/2} = nE_0$$

$$= E_{\max}$$

If $n=2$

$$|E_T| = 2E_0 \cos(\psi/2)$$

Theorem of Multiplication of Patterns (Symm. Arrays)



Step-1:-

Identify the two points that symm. makes the group and Identify Eu unit pattern.

UNIT:-

$$d = \frac{\lambda}{2}, \quad \lambda = 0$$

2-Element Array

$$E_u = \cos\left(\frac{\pi}{2} \cos\theta\right)$$

Step 2:-

Identify the group formed from the two points and its pattern

GROUP:-

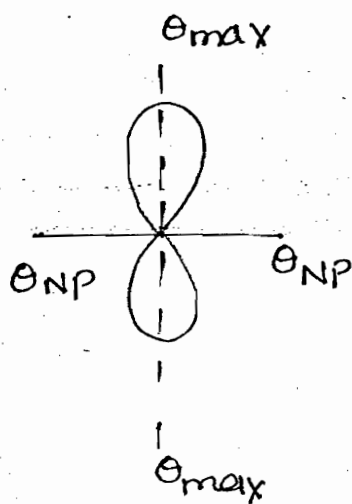
$$d = \lambda, \quad \lambda = 0$$

$$E_g = \cos(\pi \cos\theta)$$

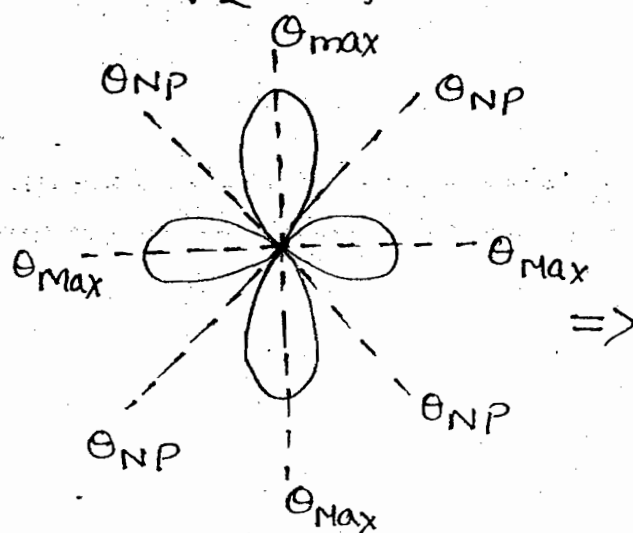
Step 3:-

Resultant Pattern = Unit Pattern $\times$ Group Pattern

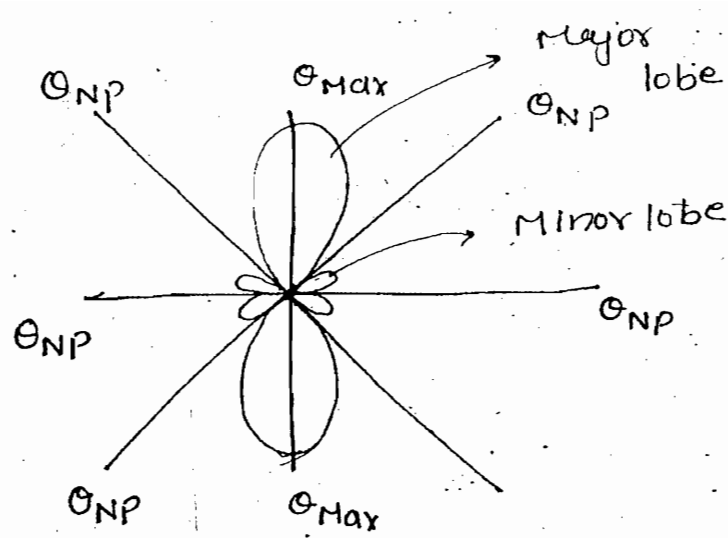
$$= \cos\left(\frac{\pi}{2} \cos\theta\right) \cos(\pi \cos\theta)$$



$\times$



⇒



Workbook! -

19. c

20. Unit - Dipole Antenna

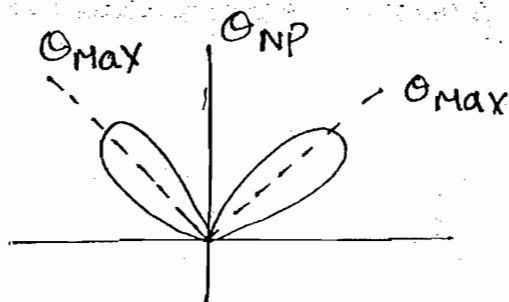
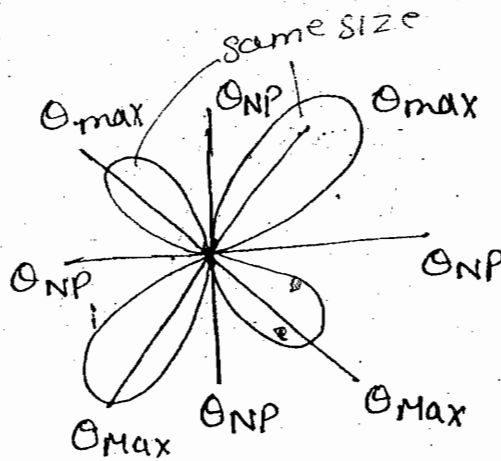
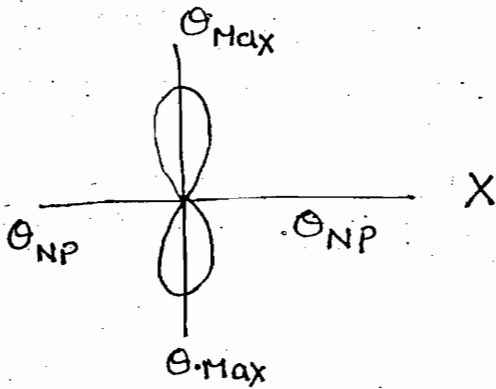
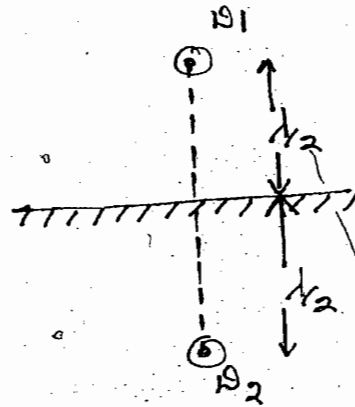
$$E_u = \cos\left(\frac{\pi}{2} \cos\theta\right)$$

GROUP →

$$d = \lambda$$

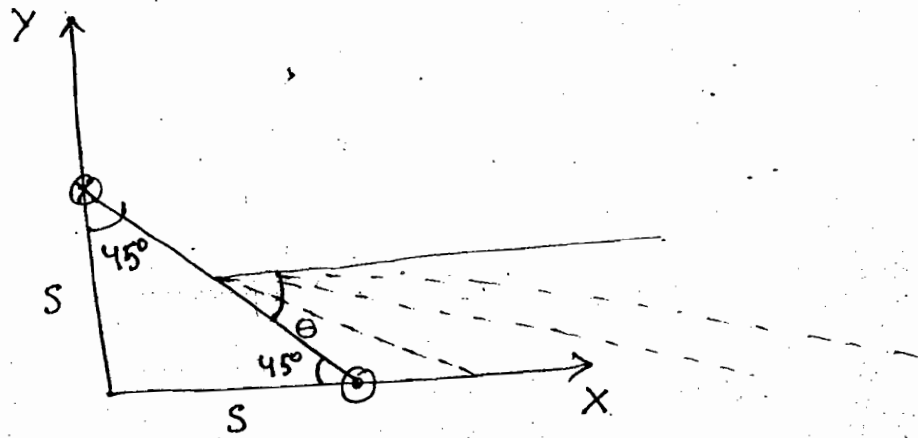
$$\lambda = \pi - 2 \text{ dipoles}$$

$$E_g = \sin(\pi \cos\theta)$$



Ans - B

$$\theta = \frac{\pi}{2} \text{ Plane} \xrightarrow{\quad} z = 0 \xrightarrow{\quad} XY \text{ Plane}$$



$$d = \sqrt{2}S, \quad \alpha = \pi$$

$$\theta = 45^\circ$$

$$\begin{aligned} \frac{E_T}{E_0} &= 2 \cos \left(\frac{\alpha + \beta d \cos \theta}{2} \right) \\ &= 2 \cos \left(\pi + \frac{2\pi}{\lambda} \cdot \frac{1}{2} \cdot \frac{1}{\sqrt{2}} \right) \\ &= 2 \sin \left(\frac{\pi S}{\lambda} \right) \end{aligned}$$

2. $\theta_{\max} = 60^\circ$ off end-fire

$$\alpha + \beta d \cos 60^\circ = 0$$

$$\alpha = -\frac{2\pi}{\lambda} \cdot \frac{1}{2} \cdot \frac{1}{2} = -\frac{\pi}{4}$$

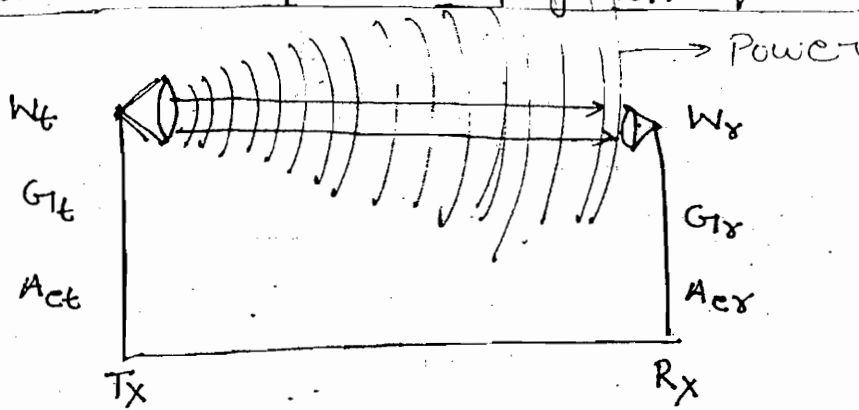
23. Unambiguous

direction finding

→ 4 lobe pattern

$$d = \lambda, \quad \alpha = 0^\circ$$

FRIIS - Free Space Propagation Equation:-



$$\text{Power density at } R_x = \frac{W_t \cdot G_t}{4\pi d^2}$$

$$W_r \text{ at } R_x = \frac{W_t \cdot G_t \cdot A_{ex}}{4\pi d^2}$$

$$A_{ex} = \frac{\lambda^2 G_r}{4\pi}$$

$$W_r = \frac{W_t \cdot G_t \cdot G_r}{\left(\frac{4\pi d}{\lambda}\right)^2}$$

$$\left(\frac{4\pi d}{\lambda}\right)^2 = L_s = \text{Loss due to spatial dispersion}$$

$$W_r (\text{dBW}) = W_t (\text{dBW}) + G_t (\text{dB}) + G_r (\text{dB}) - L_s (\text{dB})$$

$$\frac{W_t G_t}{4\pi d^2} = \frac{1}{2} \frac{E_0^2}{\eta} = \frac{E_{rms}^2}{120\pi}$$

$$E_{rms} = \frac{\sqrt{30 W_t \cdot G_t}}{d}$$

Workbook :-

17.

$$W_r (\text{dBW}) = 10 \text{ dBW} + 10 \text{ dB} + G_r (\text{dB}) - 100 \text{ dB}$$

T_x - circularly polarized

R_x - Linearly polarised.

$\frac{1}{2}$ - Power received

due to polarization mismatch

$$G_r = \frac{1}{2} \quad 10 \log \frac{1}{2} = -3 \text{ dB}$$

$$G_t = 2 = 3 \text{ dB}$$

→ doubled

24.

$$W_r = \frac{W_t \cdot G_t \cdot A_{er}}{4\pi d^2} = 0.8 \text{ W}$$

25.

$$W_r = \frac{W_t \cdot (G_t)^2}{\left(\frac{4\pi d}{\lambda}\right)^2}$$

$$d = 30 \text{ km}$$

$$\lambda = \frac{3 \times 10^8}{1 \times 10^9} = 0.3 \text{ m}$$

$$W_r = -30 \text{ dBm} \quad \text{for } W_t = 1 \text{ W}$$

$$\left(\frac{W_r}{W_t} = -30 \text{ dBm} \right)$$

$$\left(\frac{W_r}{W_t} = -30 \text{ dB} \right) \times 10^{-3}$$

$$\left(10 \log \frac{W_r}{W_t} = -30 \right) \times 10^{-3}$$

$$\left(\frac{W_r}{W_t} = 10^{-3} \right) \cdot 10^{-3}$$

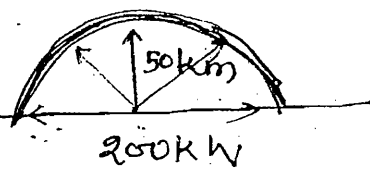
Ans - 2

$$\frac{W_r}{W_t} = 10^{-6}$$

$$E_{rms} = \frac{\sqrt{30 \cdot 1000 \cdot 1.63}}{10 \times 10^3}$$

$$P_{avg} = \frac{W_t}{2\pi d^2} a_r$$

$$= \frac{200 \times 10^3}{2\pi \times (50 \times 10^3)^2} \cdot a_r$$



→ B

$$E_{rms} \text{ --- } 5 \text{ kms --- } d$$

$$E'_{rms} \text{ --- } \frac{E_{rms}}{\sqrt{2}} \text{ --- } d' = ?$$

Further more ?

to detect 3dB decrease

$$3\text{dB in power} \text{ --- } \frac{1}{2} \text{ --- power}$$

$$\text{--- } \frac{1}{\sqrt{2}} \text{ --- field}$$

$$E \propto \frac{1}{d}$$

$$E' = \frac{E}{\sqrt{2}}$$

$$d' = \sqrt{2}d = \sqrt{2} \times 5 \text{ kms} = 7 \text{ km}$$

Further more = 2 km

$$E \propto \frac{1}{d}$$

Conventional:-

$$d = 10 \text{ cm}$$

$$f = 60 \text{ MHz}$$

$$\lambda = \frac{3 \times 10^8}{60 \times 10^6}$$

$$\approx 5 \text{ m}$$

$$d = \frac{\lambda}{50} \rightarrow \text{Hertzian dipole}$$

$$R_r = 80 \pi^2 \left(\frac{d}{\lambda} \right)^2$$

$$W_r = I_{rms}^2 R_r \quad \text{--- (1)}$$

$$I_{rms} = \frac{10 \text{ mA}}{\sqrt{2}}$$

$$\text{Efficiency} = \frac{R_r}{R_r + 0.1}$$

$$E_{rms} = \frac{\overset{\text{milli}}{1 \text{ mV}}}{\underset{\text{meters}}{d}} = \frac{\sqrt{30 \cdot W_t \cdot G_t}}{d}$$

$$\Rightarrow 10^{-3} = \frac{\sqrt{30 \cdot W_t \cdot 1.53}}{d}$$

At $\frac{\lambda}{50}$ lengths both monopole and dipole have same effect

$$\rightarrow f = 1500 \text{ MHz} \quad d = 10 \text{ cm}$$

$$\lambda = 20 \text{ cm}$$

$$d = \frac{\lambda}{2} \rightarrow \text{half wave dipole}$$

$$R_r = 73$$

$$E_{rms} = \frac{\sqrt{30 \cdot W_t \cdot 1.63}}{d}$$

$$\downarrow = ?$$